PAPER ID-310395

Subject Code: BMC105

Roll No: | | | | | | | | | | | | | |

**MCA**

**(SEM I) THEORY EXAMINATION 2025-26**

**COMPUTER ORGANIZATION & ARCHITECTURE**

**TIME: 3 HRS** **M.MARKS: 70**

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

## SECTION A

**1. Attempt *all* questions in brief.** **02 x 7 = 14**

| Q no. | Question | CO | Level |
|---|---|---|---|
| a. | List the functional units of a digital system. | 1 | K2 |
| b. | What is a bus cycle? Give one example. | 1 | K3 |
| c. | Define Booth's algorithm in one line. | 1 | K2 |
| d. | What is an instruction format? | 2 | K2 |
| e. | Define 2D memory organization. | 3 | K2 |
| f. | What is programmed I/O? | 4 | K1 |
| g. | What is SIMD also Write the full form of SIMD. | 5 | K1 |

## SECTION B

**2. Attempt any *three* of the following:** **07 x 3 = 21**

| | | | |
|---|---|---|---|
| a. | Explain signed-operand multiplication. Using Booth's algorithm, multiply –13 × +11 and show all steps. | 1 | K3 |
| b. | A memory system contains 128K words and a cache of 1K blocks, each block holding 8 words. For direct mapping, determine:<br>(i) Number of main memory blocks<br>(ii) Word offset bits<br>(iii) Index bits<br>(iv) Tag bits<br>Draw the complete physical address format. | 3 | K3 |
| c. | Explain instruction cycle with neat diagrams. Write micro-operations for Fetch, Decode, and Execute phases. | 2 | K3 |
| d. | Describe Flynn's classification. Compare SISD, SIMD, MISD, MIMD with suitable examples. | 5 | K2 |
| e. | Explain DMA mechanism. Draw the DMA transfer timing for cycle stealing, and justify why CPU performance is affected. | 4 | K3 |

## SECTION C

**3. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Design the architecture of a 3-bus CPU organization and illustrate how the instruction R1 ← R2 + (M[PC]) is executed using register transfer notation. | 1 | K3 |
| b. | Explain the following addressing modes with examples:<br>(i) Immediate<br>(ii) Direct<br>(iii) Register indirect<br>(iv) Indexed | 1 | K2 |

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**4. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Design a 4-bit Carry Look-Ahead Adder and compute:<br>$A = 1101_2$, $B = 0101_2$<br>Show all intermediate steps for carry generation and sum evaluation. | 1 | K3 |
| b. | Convert the decimal value –19.375 into IEEE-754 single precision. Show sign, exponent, mantissa, normalization, and biasing. | 1 | K3 |

**5. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Describe microprogram sequencing. Explain how conditional branching is handled in a microprogrammed control unit using a sequencer diagram. | 2 | K4 |
| b. | Explain pipelining. For a 5-stage pipeline, show pipeline execution of the instruction sequence:<br>LOAD, ADD, SUB, JUMP<br>Indicate hazards and state one solution. | 2 | K3 |

**6. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Explain 2½-D memory organization with a neat diagram. How does it improve access time compared to 2D organization? | 3 | K3 |
| b. | Define virtual memory. Explain page fault, working set, and thrashing with examples. | 3 | K3 |

**7. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Explain interrupt hardware and interrupt nesting. Draw the sequence of operations during nested interrupt handling. | 4 | K3 |
| b. | Explain parallel processing. Describe shared-memory MIMD and distributed-memory MIMD architectures with diagrams. | 5 | K2 |

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